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CT-367 TPL Assignment-01

Q1

```
/* {
```

```
#include <string.h>
```

```
int chars=0, words=0, digits=0, lines=0, spaces=0, tabs=0,
```

```
special=0;
/* }
```

```
/* /*
```

```
\n
```

```
[ ]
```

```
\t
```

```
[0-9]
```

```
[A-Za-z]
```

```
.
```

```
/* /*
```

```
int
```

```
yywrap
```

```
( )
```

```
{
```

```
return
```

```
1;
```

```
}
```

```
{ lines++; chars++; }
```

```
{ spaces++ ; chars++ ; }
```

```
{ tabs++ ; chars++ ; }
```

```
{ digits++ ; chars++ ; }
```

```
{ words++ ; chars += yyleng ; }
```

```
{ special++ ; chars++ ; }
```

```
int main (int argc, char *argv[]) {
```

```
if (argc != 2) {
```

```
printf("Clear arguments not provided\n");
```

```
return 1;
```

```
if (strcmp(argv[1], "extension.txt") != 0) {  
    printf("Error: only extension.txt is allowed\n");  
    return 1;  
}
```

```
FILE *fp = fopen(argv[1], "r");
```

```
if (!fp) {
```

```
    printf("Cannot open file %s\n", argv[1]);
```

```
    return 1;  
}
```

```
yyin = fp;
```

```
yylex();
```

```
if (chars > 0) lines++; // For 1st line
```

```
printf("\n\t File Statistics\n");
```

```
printf("Characters: %d\n", chars);
```

```
printf("Words: %d\n", words);
```

```
printf("Digits: %d\n", digits);
```

```
printf("Lines: %d\n", lines);
```

```
printf("Spaces: %d\n", spaces);
```

```
printf("Tabs: %d\n", tabs);
```

```
printf("Special characters: %d\n", special);
```

```
fclose(fp);
```

```
return 0;  
}
```

Q2

% { % }

%. %

EMAIL [A-Za-z0-9.-_+@][A-Za-z0-9.-_+@].
[A-Za-z]{2,}

DOB {(0[1-9]|1[0-9]|2[0-9]|3[0-9])\/(0[1-9]|1[0-2])\/(0[1-9]|1[0-9]|2[0-9])}

NIC [1-9][0-9]{4}-[0-9]{7}-[0-9]

%. %

{EMAIL} {printf("Email Address: %s\n", yytext);}

{DOB} {printf("Date of Birth: %s\n", yytext);}

{NIC} {printf("CNIC Number: %s\n", yytext);}

%. %

int yywrap() {return 1;}

int main() {

FILE *fp = fopen(argv[1], "r");

if (!fp) {

printf("Unable to open file %s\n", argv[1]);

return 1;

}

yyin = fp;

yylex();

fclose(fp);

return 0;

}

Q 3

$$\begin{aligned}
 \text{I)} & \text{ OR TRUE (AND TRUE FALSE)} \\
 &= \text{OR TRUE } [(\lambda p \lambda q p q p) \text{ TRUE FALSE}] \\
 &= \text{OR TRUE } [(\lambda q \text{ TRUE } q \text{ TRUE}) \text{ FALSE}] \\
 &= \text{ " " (TRUE FALSE TRUE)} \\
 &= \text{ " " } [(\lambda x \lambda y x) \text{ FALSE TRUE}] \\
 &= \text{ " " } (\lambda y \text{ FALSE}) \text{ TRUE} \\
 &= \text{OR TRUE FALSE} \\
 &= (\lambda p \lambda q p q p) \text{ TRUE FALSE} \\
 &= (\lambda q \text{ TRUE TRUE } q) \text{ FALSE} \\
 &= \text{TRUE TRUE FALSE} \\
 &= (\lambda x \lambda y x) \text{ TRUE FALSE} \\
 &= (\lambda y \text{ TRUE}) \text{ FALSE} \\
 &= \text{TRUE}
 \end{aligned}$$

$$\begin{aligned}
 \text{II)} & \text{ AND TRUE (NOT FALSE)} \\
 &= \text{AND TRUE } [(\lambda p \lambda a \lambda b p b a) \text{ FALSE}] \\
 &= \text{AND TRUE } [(\lambda a \lambda b \text{ FALSE } b a)] \\
 &= \text{ " " } [\{\lambda a \lambda b (\lambda x \lambda y y) b a\}] \\
 &= \text{ " " } [\{\lambda a \lambda b (\lambda y y) a\}] \\
 &= \text{ " " } [\{\lambda a \lambda b a\}] \\
 &= \text{AND TRUE TRUE} \quad \because \lambda x \lambda y x \Rightarrow \text{TRUE}
 \end{aligned}$$

$$\begin{aligned}
&= \text{AND TRUE TRUE} \\
&= (\lambda p. \lambda q. p \ q, p) \text{ TRUE } \text{TRUE} \\
&= (\lambda q. \text{TRUE } q, \text{TRUE}) \text{ TRUE} \\
&= \text{TRUE TRUE TRUE} \\
&= (\lambda x. \lambda y. x) \text{ TRUE TRUE} \\
&= (\lambda y. \text{TRUE}) \text{ TRUE} \Rightarrow \text{TRUE}
\end{aligned}$$

III $(+ (* 1 3) (+ 2 1))$

~~MULT~~ (P PLUS (MULT 1 3) (PLUS 2 1)) - (a)

$$\text{MULT } 1 \ 3 = [\lambda m \ \lambda n \ \lambda f \ m \ (n \ f)] \ 1 \ 3$$

$$= [\lambda n \ \lambda f \ 1 \ (n \ f)] \ 3 \Rightarrow \lambda f \ 1 \ (3 \ f)$$

$$= \lambda f \ 1 \ (\lambda f \ \lambda x \ \text{fff } x) (f)$$

$$= \lambda f \ 1 \ (\lambda x \ \text{fff } x)$$

$$= \lambda f \ (\lambda f \ \lambda x \ f \ x) (\lambda x \ \text{fff } x)$$

$$= \lambda f \ [\lambda x \ (\lambda x \ \text{fff } x) \ x]$$

$$= \lambda f \ \lambda x \ \text{fff } x = 3 \quad \text{--- (i)}$$

Now solving: -

$$\text{PLUS } 2 \ 1 = [\lambda m \ \lambda n \ \lambda f \ \lambda x \ m \ f \ (n \ f \ x)] \ 2 \ 1$$

$$= [\lambda n \ \lambda f \ \lambda x \ 2 \ f \ (n \ f \ x)] \ 1$$

$$= \lambda f \ \lambda x \ 2 \ f \ (1 \ f \ x)$$

$$= \lambda f \ \lambda x \ 2 \ f \ \text{fff } x = 23 \quad \text{--- (ii)}$$

Substitute eq (i), (ii) in (a)

$$= \text{PLUS } 3 \ 3$$

$$= [\lambda m \lambda n \lambda f \lambda x \ m \ f \ (n \ f \ x)] \ 3 \ 3$$

$$= [\lambda n \lambda f \lambda x \ 3 \ f \ (n \ f \ x)] \ 3$$

$$= \lambda f \lambda x \ 3 \ f \ (3 \ f \ x) = \lambda f \lambda x \ 3 \ f \ (f \ f \ f \ x)$$

$$= \lambda f \lambda x \ f \ f \ f \ f \ f \ x = 6$$

$$\text{IV) } (\gamma \text{ fact}) \ 2 = 2 \text{ --- (a)}$$

$$= [\lambda f \{ \lambda x \ f \ (x \ x) \} \{ \lambda x \ f \ (x \ x) \}] \text{ fact } 2$$

$$= [\cancel{\{ \lambda x \ f \ (x \ x) \}}] [\{ \lambda x \ \text{fact} \ (x \ x) \} \{ \lambda x \ \text{fact} \ (x \ x) \}]$$

$$\text{Suppose } B = \lambda x \ \text{fact} \ (x \ x) \text{ --- 2nd term}$$

$$\Rightarrow \{ \lambda x \ \text{fact} \ (x \ x) \} \ B \ 2$$

$$\Rightarrow \text{fact} \ (B \ B) \ 2$$

So from this, we see that

$$= \text{fact} [\{ \lambda x \ \text{fact} \ (x \ x) \} \{ \lambda x \ \text{fact} \ (x \ x) \}] \ 2$$

Now from here we see that inner term is repeated
So used property

$$\gamma \text{ fact} \longrightarrow \text{fact} \ (\gamma \text{ fact})$$

on eq (a)

Now

$$\Rightarrow (\text{fact } (\lambda \text{ fact})) 2$$

$$\because \text{fact} = \lambda f \lambda n \text{ if } n=0 \text{ then } 1 \text{ else } n * (f(n-1))$$

Now for 2, 1st check:

$$\Rightarrow \text{ISZERO } 2 = [\lambda n n (\lambda x \text{ FALSE}) \text{ TRUE}] 2$$

$$= 2 (\lambda x \text{ FALSE}) \text{ TRUE}$$

$$= (\lambda f \lambda x f x) (\lambda x \text{ FALSE}) \text{ TRUE}$$

$$= (\lambda f \lambda x f x) \text{ FALSE} [\lambda f \lambda x \{ \lambda x \text{ FALSE} \} \{ \lambda x \text{ FALSE} \} x]$$

$$= (\lambda x \text{ FALSE}) (\lambda x \text{ FALSE}) \text{ TRUE}$$

$$= (\lambda x \text{ FALSE}) \text{ FALSE} = \text{FALSE}$$

Since FALSE so else brach will run.

$$\Rightarrow n * (f(n-1)) (\lambda \text{ fact}) 2$$

$$= n * [(\lambda \text{ fact}) (n-1)] 2$$

$$= 2 * [(\lambda \text{ fact}) (1)] - (b)$$

Now evaluate eq (b) similtiarly

$$(\lambda \text{ fact}) 1 \Rightarrow \text{fact } (\lambda \text{ fact}) 1$$

Again check condition

$$\text{ISZERO } 1 = [\lambda n n (\lambda x \text{ FALSE}) \text{ TRUE}] 1$$

$$= 1 (\lambda x \text{ FALSE}) \text{ TRUE}$$

$$= (\lambda f \lambda x f x) (\lambda x \text{ FALSE}) \text{ TRUE}$$

$$= \{ \lambda x (\lambda x \text{ FALSE}) x \} \text{ TRUE} \Rightarrow (\lambda x \text{ FALSE}) \text{ TRUE}$$

$$= \text{FALSE}$$

So,

$$n * ((\text{fact}(n-1)) (\gamma \text{ fact}) 1) \\ \Rightarrow 1 * (\gamma \text{ fact}(0)) - (c)$$

Now evaluate $\text{fact}(0)$:-

$$\Rightarrow (\gamma \text{ fact}) 0 \Rightarrow \text{fact}(\gamma \text{ fact}) 0$$

So check condition:-

$$\text{IS ZERO } 0 = \{ \lambda n n (\lambda x \text{ FALSE}) \text{ TRUE} \} 0$$

$$= 0 (\lambda x \text{ FALSE}) \text{ TRUE}$$

$$= (\lambda f \lambda x x) (\lambda x \text{ FALSE}) \text{ TRUE} \Rightarrow (\lambda x x) \text{ TRUE}$$

$$= \text{TRUE} \quad n = 0$$

so it will execute if-statement and return 1

Now do backward substituting in eq (c), (b)

$$(\gamma \text{ fact}) 1 = 1 * 1$$

$$= (\lambda m \lambda n \lambda f m (n f)) 1 1$$

$$= (\lambda n \lambda f 1 (n f)) \Rightarrow \lambda f 1 (1 f)$$

$$= \lambda f (\lambda f \lambda x f x) (\lambda f \lambda x f x) f$$

$$= \lambda f (\lambda f \lambda x f x) (\lambda x f x) \Rightarrow \lambda f \{ \lambda x (\lambda x f x) x \}$$

$$= \lambda f \lambda x f x = 1$$

so it will return 1. Then for eq (b)

$$(\gamma \text{ fact}) 2 = 2 * 1 = (\lambda m \lambda n \lambda f m (n f)) 2 1$$

$$= \lambda f 2 (1 f) \Rightarrow \lambda f (\lambda f \lambda x f x) (\lambda f \lambda x f x) f$$

$$= \lambda f (\lambda f \lambda x f f x) (\lambda x f x) \Rightarrow \lambda f \{ \lambda x (\lambda x f x) (\lambda x f x) x \}$$

$$= \lambda f \lambda x (\lambda x f x) (f x) \Rightarrow \lambda f \lambda x f f x$$

$$(\gamma \text{ fact}) = 2 \text{ Answer}$$